

- 2 A student investigates a circuit containing capacitors. The circuit is connected with a capacitor of capacitance A , as shown in Fig. 2.1.

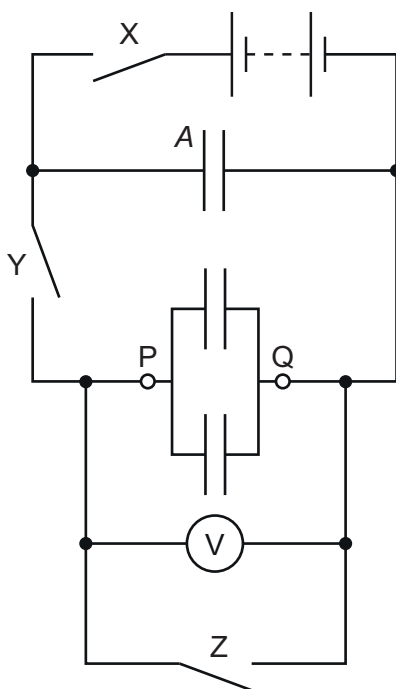


Fig. 2.1

Two capacitors, each of capacitance C , are connected in parallel between P and Q.

Initially, switch X and switch Z are closed and switch Y is open.

Switches X and Z are opened. Switch Y is then closed. The maximum potential difference between P and Q is measured using the voltmeter. This procedure is repeated and the mean maximum potential difference V between P and Q is determined.

The experiment is then repeated by changing the number n of capacitors, each of capacitance C , connected in parallel between P and Q.

It is suggested that V and n are related by the equation

$$EA = V(nC + A)$$

where E is the electromotive force (e.m.f.) of the battery.

- (a) A graph is plotted of $\frac{1}{V}$ on the y -axis against n on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

